Data Imbalance Testing and Handling System

CODE: 26-1-D-3

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What Is Data Imbalance?

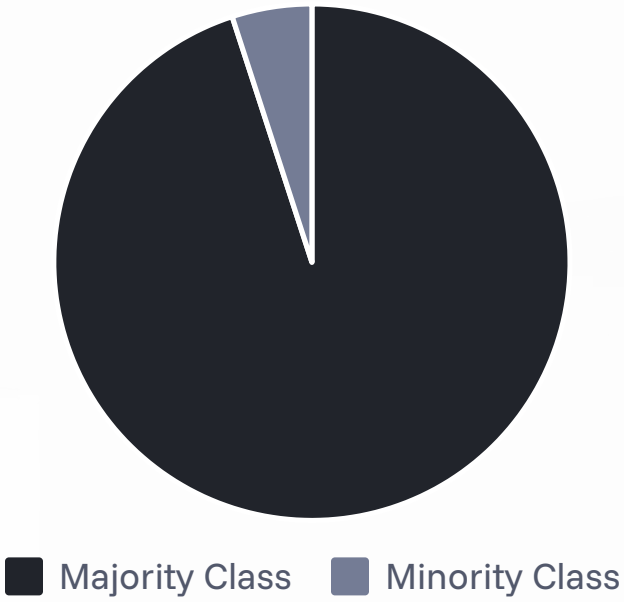
When one class appears far more often than others in the dataset.

Many real datasets are uneven: most samples belong to one "majority" class.

The minority class can be rare but important (e.g., fraud, faults, disease).

A model can look accurate by mostly predicting the majority class.

Therefore we must detect imbalance and evaluate with the right metrics.



ABSTRACT

Problem

- Real-world datasets frequently suffer from class imbalance.
- Critical minority classes are underrepresented.
- Leads to misleadingly high model accuracy.
- Masks severe failures in crucial, rare cases (fraud detection, medical diagnoses).

Challenge

- Existing solutions are fragmented.
- Poor documentation and inconsistent methodologies.
- Difficult to reproduce results.
- Hard to reliably compare different handling techniques.

Our Solution

- Unified system for diagnosing data imbalance.
- Implementing handling strategies.
- Comparing results effectively.
- Generating comprehensive documentation.

Real-World Impact

Class Imbalance

Skewed data leads to biased models.

Performance Risk

Models fail on rare, critical cases.

Consequences

Affects fraud, medical diagnosis, fairness.

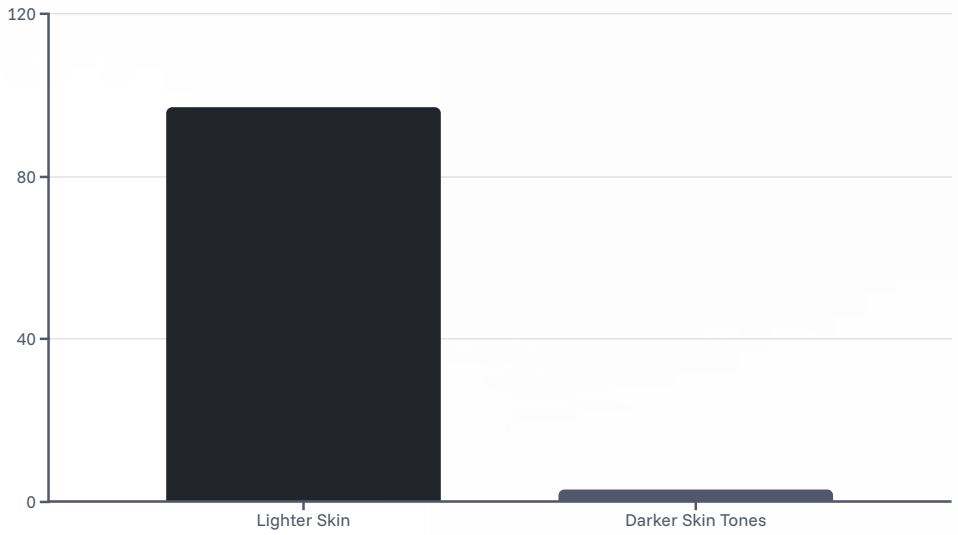
Real-World Example: Melanoma Detection Bias

- **Imbalanced medical datasets can make models less reliable for minority groups, especially in high-stakes diagnosis.**
- **In dermatology AI, severe data skew can lead to systematic bias.**
- **Models can perform well on lighter skin, while frequently failing to recognize life-threatening conditions on darker skin tones.**

97%

Melanoma prediction models at high risk of bias due to severe data skew (Kleinberg et al., 2022).

Performance by Skin Tone



The Accuracy Paradox: Why Accuracy Misleads

Consider the following confusion matrix for an imbalanced dataset:

	Predicted Positive	Predicted Negative
Actual Positive	TP: 0	FN: 50
Actual Negative	FP: 0	TN: 4950

The accuracy for this model is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{0 + 4950}{0 + 4950 + 0 + 50} = \frac{4950}{5000} = 0.99$$

- 99% accuracy achieved by ignoring minority class entirely
- Perfect accuracy on majority, complete failure on minority
- Demonstrates why accuracy alone is insufficient

Diagnosing Imbalance: Metrics & Pitfalls

Key Diagnostic Metrics

Imbalance Ratio (IR)

Measures class skew severity.

$$IR = \frac{N_{maj}}{N_{min}}$$

Higher IR = more severe imbalance.

G-Mean (Geometric Mean)

Balances sensitivity & specificity.

$$G - Mean = \sqrt{TPR \times TNR}$$

TPR = True Positive Rate (Recall/Sensitivity)

TNR = True Negative Rate (Specificity)

Low G-Mean indicates majority class bias.

IR shows how severe the class skew is, and G-Mean checks whether performance is balanced across both classes.

Without them, accuracy can look “good” while the model is biased toward the majority class.

Two Main Families of Solutions

Data-Level Approach (External Method)

Modify dataset before training to reduce class imbalance.

Algorithm-Level Approach (Internal Method)

Adjust the learning algorithm to prioritize the minority class .

Project Focus: Study both approaches for a transparent, reproducible workflow.

Data-Level Methods: Balancing Through Resampling

- **Modify dataset before training to reduce class imbalance**
- **Two main approaches: Over-sampling and Under-sampling**

Over-sampling

Increase minority class samples

Under-sampling

Reduce majority class samples

OVER-SAMPLING

SMOTE: Synthetic Minority Over-sampling

How SMOTE Works

01

Select minority sample

02

Find k-nearest neighbors

03

Generate synthetic samples via interpolation

Benefits

- Increases minority class without duplication
- Adds variation via interpolation

Risks

- Risk of overgeneralization
- Can cause class overlap

NearMiss: Distance-Based Under-Sampling

Strategically selects majority samples using distance rules.

Key Variants:

- NearMiss-1: Keeps majority samples closest to minority neighbors
- NearMiss-2: Uses average distance to farthest minority neighbors
- NearMiss-3: For each minority sample, keeps closest majority samples
- Most Distant: selects majority samples whose average distances to the three nearest minority samples are the largest

Trade-off:

- Deleting data discards potentially valuable information

Existing Tools & Software

Imbalanced-learn

- Scikit-Learn Integration
- Comprehensive Resampling Library
- High Academic Reliability

SMOTE-variants (Python Package)

- 85 over-sampling methods
- Multi-Class & Selection
- Research Specialized

WEKA

- No-Code Interface
- Dual Strategies
- Extensible

KEEL

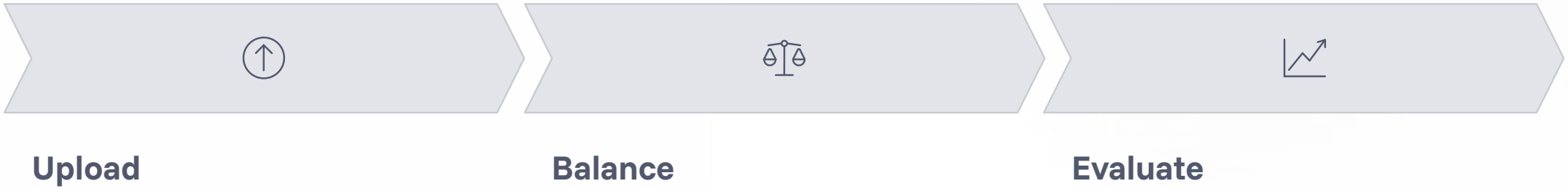
- Standardized Benchmarking Repository
- Specialized Imbalanced Module
- Scientific Validation Tools

Identified Gap: Existing tools lack integrated workflow for diagnosis, handling, evaluation, and documentation.

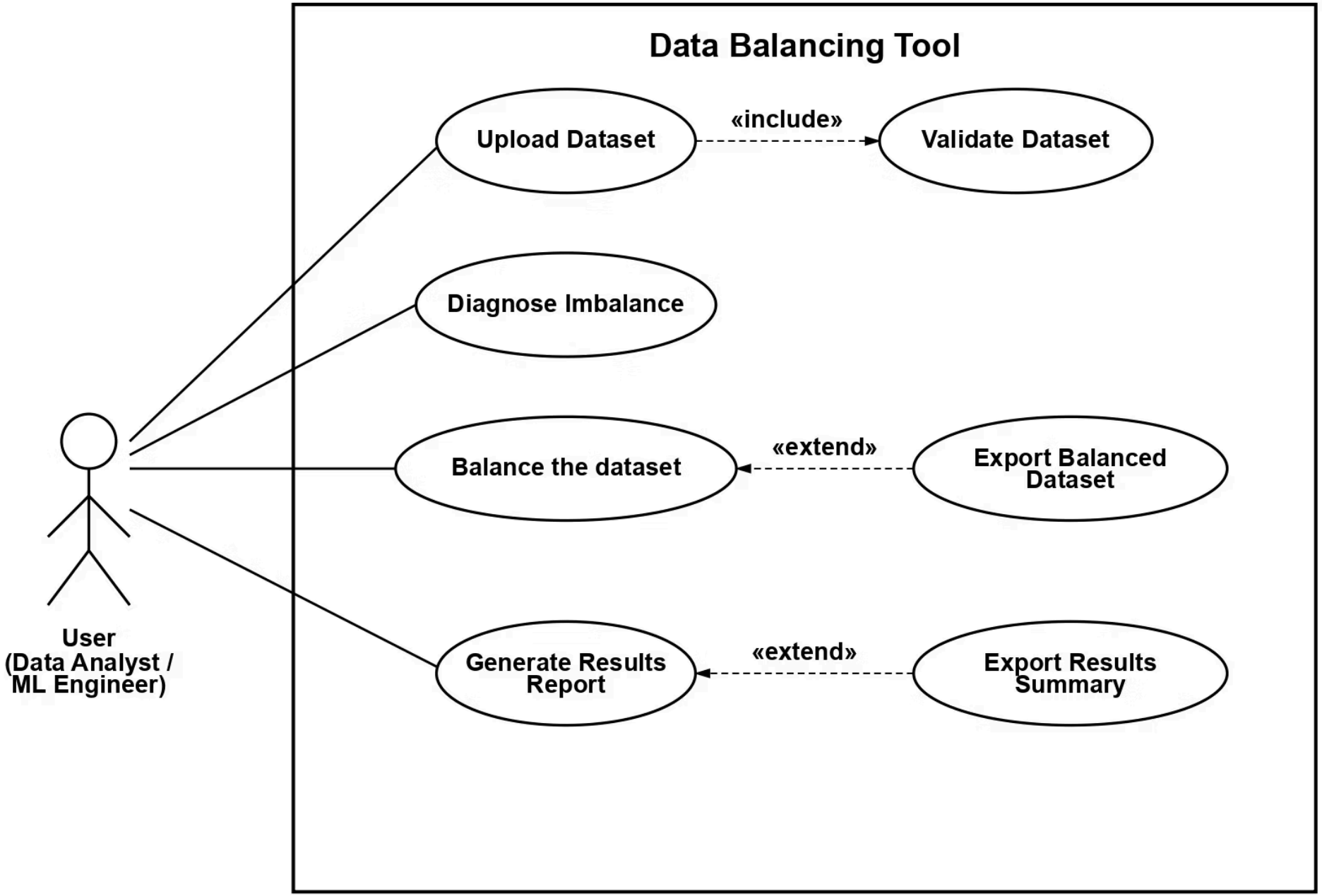
From Methods to System Usage

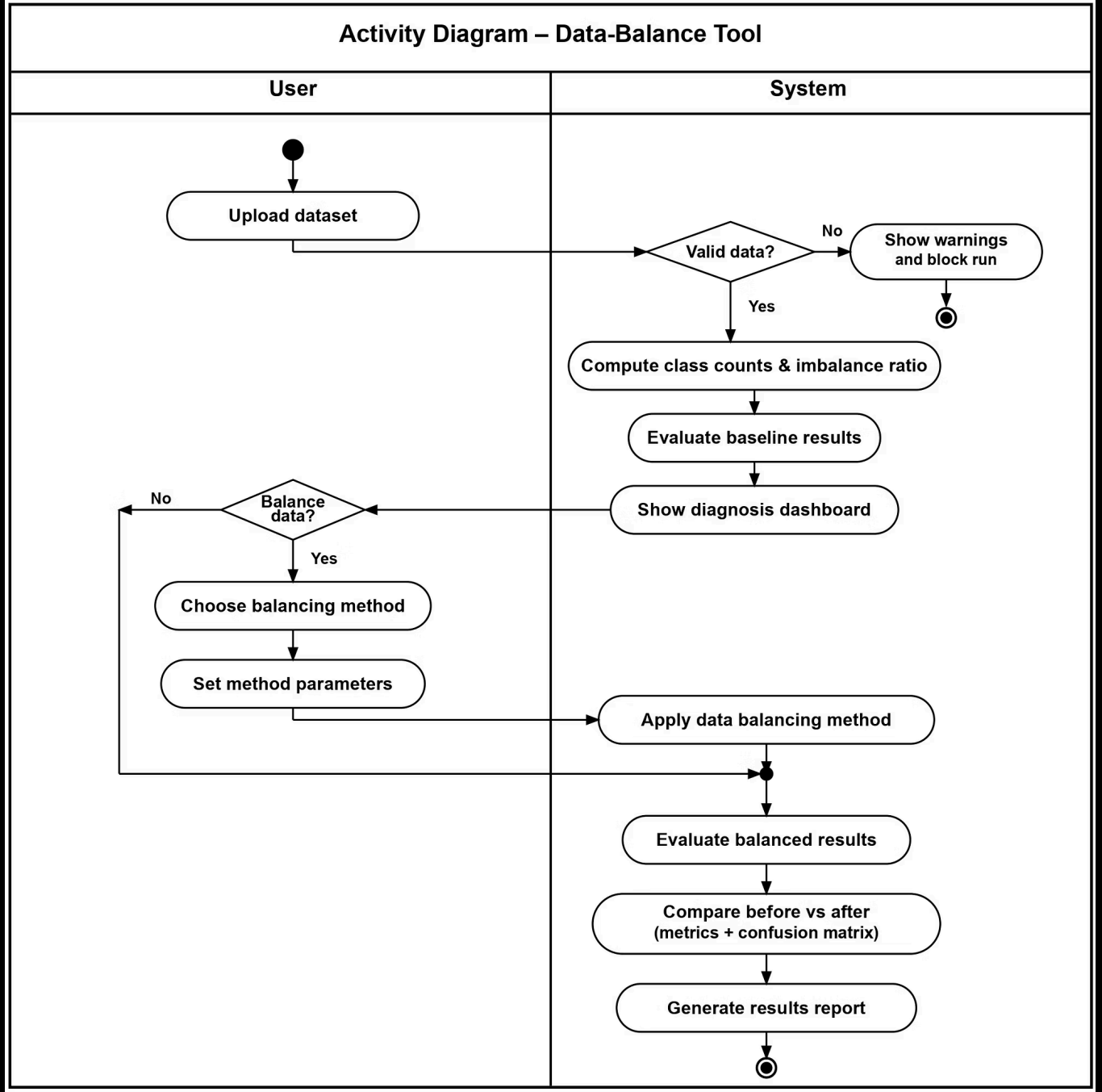
Next: who uses the system and what actions they perform (Use Case Diagram)

- Users: Researchers, Software Engineers, AI Developers.
- Core actions: Upload dataset, Analysis, Balancing method selection, Export.
- Outputs: Imbalance metrics + Evaluation, comparison report
- Goal: validate improvements without data leakage



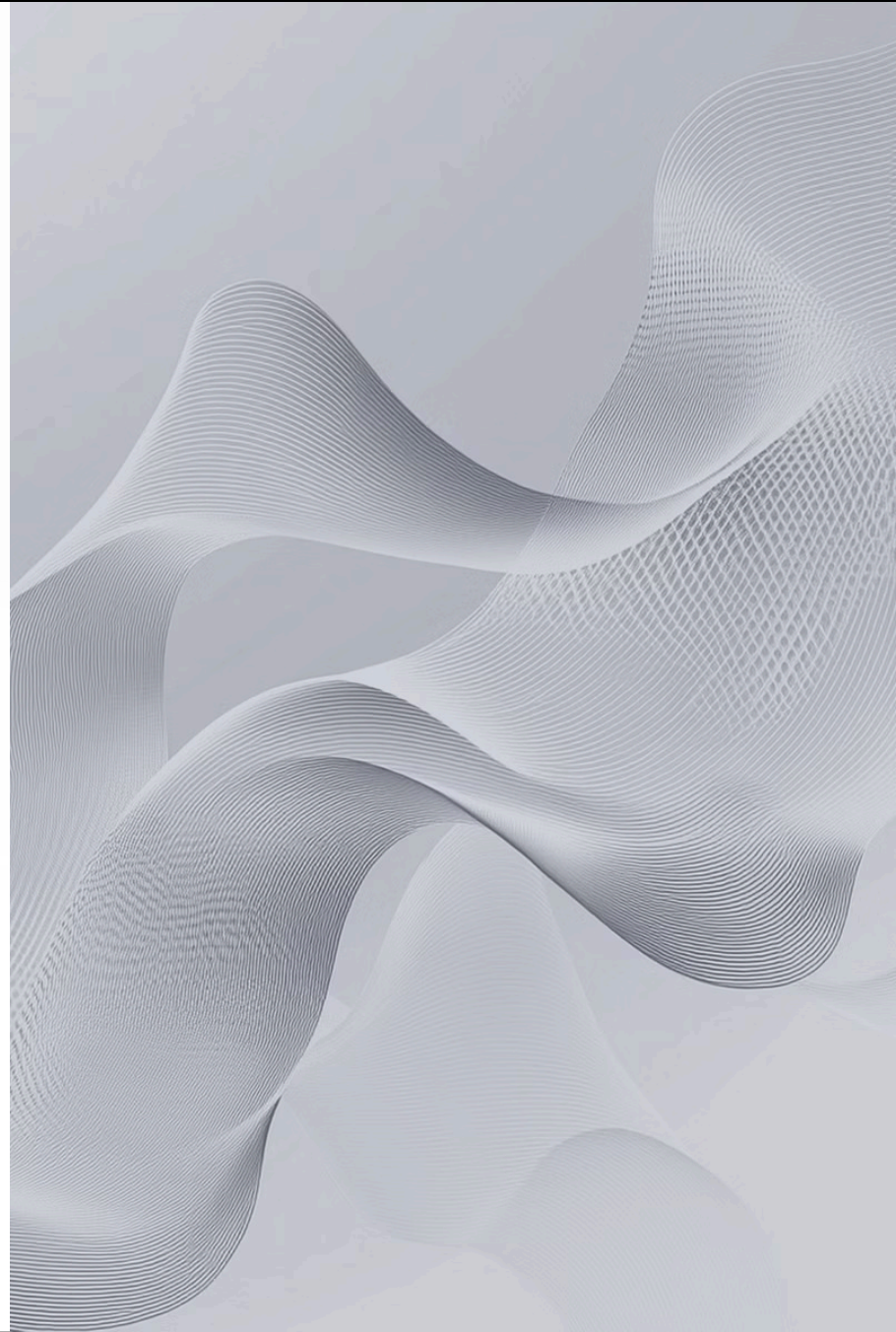
Use Case Diagram




Activity Diagram – Data-Balance Tool
User
System
Upload dataset
Valid data?
No
Show warnings and block run
Yes
Compute class counts & imbalance ratio
Evaluate baseline results
Show diagnosis dashboard
Balance data?
No
Yes
Choose balancing method
Set method parameters
Apply data balancing method
Evaluate balanced results
Compare before vs after (metrics + confusion matrix)
Generate results report

System Demo Walkthrough

The following screens showcase our system's interface and key features.



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Project Overview

Manage your dataset balancing projects

New Run

Upload a dataset and start a new balancing workflow

Start New Project

Open Previous Run

Continue working on a saved project or review past results

Browse Projects

Recent Runs

Your latest balancing projects

Dataset Name	Date	Method Used	Best Metric
credit_risk.csv	2026-01-05	SMOTE	**0.89 F1**
customer_churn.csv	2026-01-03	Hybrid	**0.84 F1**
fraud_detection.csv	2025-12-28	NearMiss	**0.76 F1**

What this tool does

This tool helps you identify and fix class imbalance in machine learning datasets to improve model performance on minority classes.

Upload your dataset, diagnose imbalance issues, apply balancing techniques like SMOTE or undersampling, and evaluate the impact on key metrics.

Export balanced datasets ready for training along with detailed reports for reproducibility.

Upload Dataset

Upload your CSV file and configure label settings

File Details

File name	Rows	Columns
credit_risk.csv	12,847	23

Replace File

Label Configuration

Select the target column for classification

Select label column

default

Choose the column that contains your classification target

Detected classes: 0 (non-default), 1 (default)

Validation Results

Dataset quality checks and warnings

- No missing values in label column — PASS
- Sufficient samples per class (min: 150) — PASS
- Severe class imbalance detected (IR: 8.4:1) — WARN
- 4 features contain missing values (5-12% per feature) — WARN

Evaluation

Compare model performance before and after balancing

Before Balancing

Classification Metrics

Accuracy	0.879
Precision	0.623
Recall	0.412
F1 Score	0.496
Macro F1	0.681

Confusion Matrix

	Pred 0	Pred 1
Actual 0	2,184	76
Actual 1	182	128

After Balancing

Classification Metrics

Accuracy	0.841
Precision	0.781
Recall	0.823
F1 Score	0.801
Macro F1	0.836

Confusion Matrix

	Pred 0	Pred 1
Actual 0	2,003	257
Actual 1	55	255

Performance Delta

Changes in key metrics after balancing

Accuracy	-0.038 ↓
Precision	+0.158 ↑
Recall	+0.411 ↑
F1 Score	+0.305 ↑
Macro F1	+0.155 ↑

Summary: Significant improvement in minority class detection (+41% recall) with strong F1 gain (+30%).

Diagnosis

Analyze class distribution and imbalance severity

Class Counts

Sample distribution per class

Class	Count
0 (non-default)	11,298
1 (default)	1,549
Total	12,847

Imbalance Ratio (IR)

Majority to minority ratio

8.4:1

Severe imbalance

IR = 11,298 / 1,549 ≈ 7.29
Rounded to 8.4:1 for display

Accuracy Warning

Why accuracy can mislead

A naive model predicting only class 0 would achieve 87.9% accuracy without learning anything useful.

Focus on Precision, Recall, and F1 scores for the minority class instead.

Class Distribution

Visual representation of imbalance

Class 0 (non-default) — 11,298 (87.9%)

Class 1 (default) — 1,549 (12.1%)

Majority Class	Minority Class	Imbalance Level
Class 0	Class 1	Severe

Continue to Balancing

Evaluation & Success

Evaluation Approach

Software Tests

- Data validation
- Algorithm correctness
- Export integrity

Research Tests

- Fair comparisons
- Reproducibility

Key Performance Indicators

- **Runtime:** <60s for supported datasets
- **Imbalance reduction:** IR decreases after balancing
- **Minority-class recall:** Improves on most datasets
- **Macro-F1:** Increases or stable
- **Per-class reporting:** 100% Precision/Recall/F1
- **Export success:** 100% in normal usage

Success Criteria

1. Accurate diagnosis of imbalance level
2. Effective balancing improves class distribution
3. Clear before/after comparison shows balancing effect

Conclusions

- Class imbalance causes models to fail on critical minority cases despite high accuracy
- Accuracy alone is misleading—must use balanced metrics (G-Mean, per-class recall, Macro-F1)
- Our unified workflow enables reproducible diagnosis, handling, and comparison
- Transparent documentation ensures reliable methodology and fair evaluation

Thank You

Questions?

GitHub repository and appendices available. Contact information available upon request.